Chap-2

Data Integrat^n & Transformat^n:

↓ merging of data from multiple data store

* Correlation relat^nship bet^n two attributes can be discovered by χ^2 (chi-square) test.

Example 2.1 মনে কর a group of 1500 people was surveyed. Gender note করা হয়েছিল। প্রত্যেক person এর বই পড়ার preference poll এর হলো — fiction না non fict^n পড়তে পছন্দ করে। এক্ষেত্রে দুটির attribute হলো ① Gender
② preferred reading

Table 2.2: A 2×2 contingency table for the data

Example 2.1
"Are gender and preferred_Reading correlated?"

	male	female	Total
fiction	250 (90)	200 (360)	450
non-fiction	50 (210)	1000 (840)	1050
	300	1200	1500

প্রত্যাশিত

N = no. of data tuples
count$(A=a_i)$ = no. of

$$e_{11} = \frac{count(A=a_i) \times count(B=b_j)}{N}$$

$$= \frac{count(male) \times count(fiction)}{N} = \frac{300 \times 450}{1500} = 90$$

expected frequency
$$\begin{cases} e_{12} = \dfrac{\overset{30}{\cancel{450}} \times 1200}{\cancel{1500}} = 360, \quad e_{21} = \dfrac{\overset{70}{\cancel{1050}} \times \cancel{300}}{\cancel{1500}} = 210 \\ e_{22} = \dfrac{\overset{210}{\cancel{1050}} \times \overset{4}{\cancel{1200}}}{\overset{5}{\cancel{1500}}} = 840 \end{cases}$$

sum of expected frequency = total observed frequency (row এর column হবে)

$$\chi^2 = \sum_{i=1}^{c} \sum_{j=1}^{r} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

$$= \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840}$$

$$= 284.44 + 121.90 + 71.11 + 30.48 = 507.93$$

For 2×2 table, the degree of freedom are $(2-1)(2-1)=1$.

For 1 degree of freedom, the χ^2 value needed to reject the hypothesis at the 0.001 significance level is 10.828 — Statistics এর text book (২৭৮ পৃষ্ঠা)

$\because$ আমাদের computed value $507.93 > 10.828$ সেহেতু We can reject the hypothesis "gender & preferred reading are independent"

সুতরাং, আমরা বলতে পারি two attributes are strongly correlated for the given group of people